

Students Name:	Grade: 6	Date:
Subject: Mathematics	Unit Title: Numerical Reasoning	

Question-1

a) Welcome to the World of Science and Numbers Festival!

The festival organisers have found an old record book, but the category labels have been smudged. Your job as a Junior Number Detective is to identify the correct type of number for each entry before the festival opens.

Identify the most appropriate form of number for each value. Present your answer in a table.

Festival Record	Value	Suitable Form of Number
Number of stalls at the festival	120	
Empty parking spaces	0	
Temperature at the night show	-5°C	
Weight of prize trophy	2.5 kg	
Entry fee discount	25%	

b) The festival ticketing system must calculate the following expression before printing. Using the correct order of operations (BODMAS), **Calculate** $96 \div 8 + 6 \times 4 - 13$.

c) The festival noticeboard displays some mathematical statements. **Recognise** the property shown in each statement.

- i) $7 \times 1 =$ _____
- ii) $4 \times (3 + 6) = 4 \times 3 + 4 \times 6$ _____
- iii) $14 + 0 =$ _____
- iv) $7 + 9 = 9 + 7$ _____

Question 2:

Calculate and complete the table below by filling in the missing decimal, fraction, or percentage for each row. **Write down** all fractions in their simplest form.

Decimal	Fraction	Percentage	Rough Work
0.45			

	$\frac{2}{5}$		
0.875			
	$\frac{3}{25}$		
0.64			
	$\frac{1}{8}$		
0.015			
	$\frac{3}{4}$		
0.4			
	$\frac{6}{20}$		

Question 3:

The festival food court sells snacks to visitors from around the world. A family orders the following items.

Item	Price	Rough Work
Fruit Salad	₹72.50	
Fresh Juice	₹48.25	
Sandwich	₹34.75	

The festival café is running a special promotion and adds a 20% service charge on the total bill.

a) **Calculate** the total cost of all three items before the service charge is added.

b) **Determine** the final amount the family needs to pay after the 20% service charge is applied.

c) The festival baker uses two different flour measurements for a recipe. The first measurement is $\frac{5}{6}$ kg of flour. The second measurement is $\frac{1}{3}$ kg. **Simplify** $\frac{5}{6} - \frac{1}{3}$. Give your answer in its simplest form.

Question 4:

The festival committee is setting up its main performance stage.

There are 60 spotlights and 90 speaker boxes.

The stage manager wants to arrange them in equal groups so that no item is left over.

a) Using prime factorisation, **Determine** the Highest Common Factor (HCF) and Lowest Common Multiple (LCM) of 60 and 90.

- b) The stage manager says, "Prime factorisation makes it easier to find the LCM." **Explain** why this method is useful.
- c) A festival souvenir was bought for ₹1,800 and later sold for ₹1,350. **Determine** the loss and the loss percentage.

Question 5:

The festival is creating a star display pattern using LED tiles. The first few arrangements are shown below.

Display Number	Number of Stars
1	5
2	10
3	15
4	
5	
6	

- a) Complete the table by finding the number of stars needed for Displays 5 and 6.
- b) Identify what pattern you notice in the table?
- c) State the number of stars needed for Display 12.

Question 6:

The festival sells hand-made bracelets for ₹120 each. The table below shows the total money collected.

Number of Bracelets	Money Collected (₹)
3	360
6	720
9	1,080
12	
15	
18	
21	

a) Complete the table.

b) Describe the pattern of how the money collected changes as the number of bracelets increases.

Question 7:

The festival has set up an ancient sand clock (hourglass) as an exhibit. Every minute, the clock drops 20 grams of sand. The observations are shown below.

The festival wants a rule that can be used for any number of minutes

Minutes (m)	Sand Dropped (g)
1	20
2	40
3	60
4	80
6	120

a) **Develop** a mathematical rule connecting Minutes (m) and Sand Dropped (s).

b) **Generalise** your rule using variables

c) **Verify** whether your rule works for 15 minutes. **Show** your working.

Question 8:

The festival director wants to use your rule for a new exhibit. A new sand clock is designed. It drops 20 grams every minute. A student makes the following claim.

"If the number of minutes triples, the amount of sand dropped will always triple."

a) **Apply** your rule from Question 7 to calculate the amount of sand dropped after 24 minutes.

b) **Verify** whether the student's claim is correct by testing it with 5 minutes and 15 minutes.

c) **Evaluate** whether the claim will always be true. **Justify** your answer using mathematical evidence from the pattern.